

Programming and Data Structures Laboratory (CS19003)

Assignment 4 (Section 6): 1D array

Department of CSE, IIT Kharagpur

4th February, 2026

General instructions to be followed strictly:

- Name your file as `<roll_no>_<assignment_no>_<question_no>`. For example, if your roll number is 14CS10001 and you are submitting assignment 3 question number (a), then name your file as 14CS10001_3-a.c
- Write your name, roll number, and assignment number at the beginning of your program.
- Use proper indentation in your code and comment. Marks may be deducted without proper indentation and comments.
- Submitting the **correct file** is your responsibility, any request of re-submission after the lab will **never** be entertained.
- **Zero tolerance** policy will be strictly followed for **plagiarism**. The marks will be put straight zero if such cases are identified.

Question-1 (15 points)

You are given a one-dimensional array of n positive integers, where $n \geq 2$. Your task is to examine all possible pairs of distinct elements in the array and determine the pair whose *Greatest Common Divisor* (GCD) is the maximum.

The GCD of two positive integers is defined as the largest positive integer that divides both numbers exactly. You must compute the GCD using only loops and conditional statements, without using any built-in library functions, recursion, pointers, or user-defined functions. If more than one pair has the same maximum GCD, consider the first such pair encountered while traversing the array from left to right. Print the two elements forming the pair and the corresponding maximum GCD.

Example:

(Input)

Enter the number of elements in the array: 5

Enter the array elements: 12 18 24 30 9

(Output)

Pair: 12 24

Maximum GCD: 12

Question-2 (20 points)

You are given a one-dimensional array of n positive integers. A *divisible subarray* is defined as a contiguous sequence of elements such that each element (except the first) is divisible by the element immediately preceding it.

Your task is to determine the length of the longest contiguous subarray that satisfies this divisibility condition. The subarray must consist of consecutive positions in the array, and only adjacent elements should be considered. Print the length of the longest such subarray. If no such subarray exists, print 0.

Example:

(Input)

Enter the number of elements in the array: 6

Enter the array elements: 2 4 8 3 6 12

(Output)

The longest subarray length is 3

Question-3 (15 points)

You are given a one-dimensional array of n integers where n is an odd integer. First, count the number of even elements and the number of odd elements present in the array.

If the number of even elements is greater than the number of odd elements, rotate the array to the *left by one position*. Otherwise, rotate the array to the *right by one position*. A left rotation shifts each element one position to the left, moving the first element to the last position. A right rotation shifts each element one position to the right, moving the last element to the first position. After performing the required rotation, print the modified array.

Example:

(Input)

Enter the number of elements in the array: 7

Enter the array elements: 4 8 3 6 2 9 1

(Output)

Modified array: 8 3 6 2 9 1 4